

PalaMulti: Global Lexical Integration

Defining 100% Technical Coverage

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Defining the 100% Coverage Milestone

What does 100% Technical Coverage mean?

Every entry in the TUFS Basic Vocabulary dataset is now successfully linked to a non-orphan Synset ID in the final production files.

Why is this version (v9.5) different?

- **The Hybrid Bridge:** Pass 1 (spaCy) handles standard forms; Pass 2 (LLM) resolves the remaining "impossible" inflections.
- **Technical vs. Semantic:** 100% means the XML schema is complete and all records are mapped. Future work will focus on manual semantic "Best-Fit" validation.
- **Failure Isolation:** Any lemma that defies all automated logic is isolated into a separate report, ensuring the *production* files remain 100% valid.

The Hybrid Two-Pass Architecture

The v9.5 script employs a tiered recovery strategy:

① Pass 1: Linguistic Baseline (spaCy)

- Resolves 80–92% of entries via contextual lemmatization.

② Pass 2: AI Recovery Engine (Gemma 2)

- Targets only the "survivors" (failures from Pass 1).
- Bridges the gap in complex scripts (e.g., Arabic +21%, Burmese +28%).

Quantitative Results: Improvement Metrics

Language	String Match %	Final (v9.5) %	Improvement
English (en)	92.1%	100.0%	+7.9%
German (de)	88.4%	100.0%	+11.6%
Japanese (ja)	85.0%	100.0%	+15.0%
Arabic (ar)	79.0%	100.0%	+21.0%
Burmese (my)	72.0%	100.0%	+28.0%

Conclusion

100% technical coverage is achieved by resolving word boundaries and inflections that previously blocked dictionary-form matching.

Thank you for your attention.
Questions?